

Chapter 38. Airway Management

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Airway Management: Introduction

“Airway management” implies the provision of assistance to a patient in maintaining a patent airway. This chapter will review airway anatomy and basic airway management techniques.

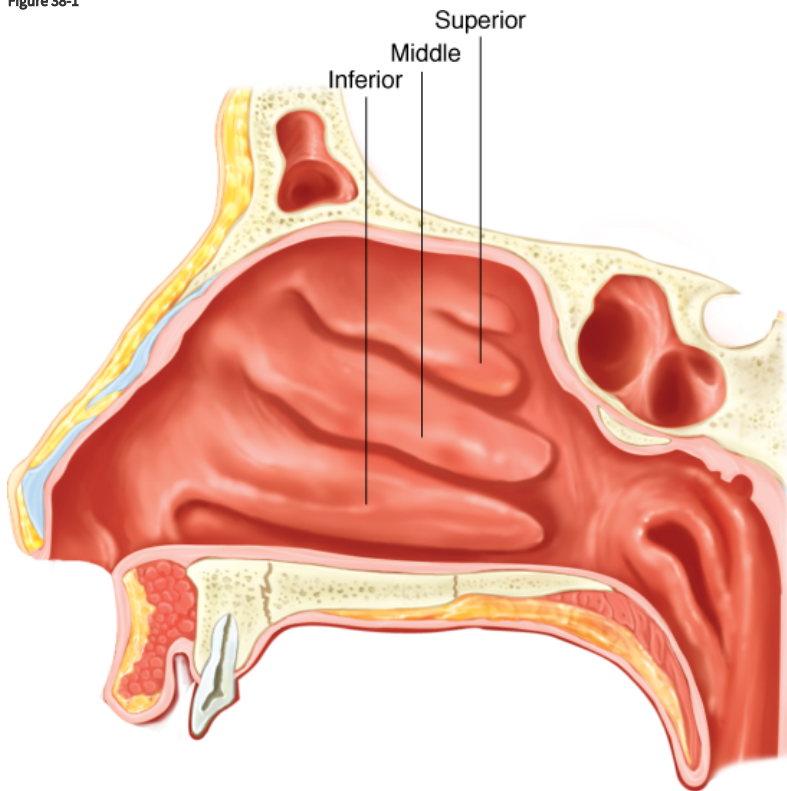
Airway Anatomy

The Nose and Nasopharynx

The nose is lined with vessel-rich mucosa designed to warm and humidify the air. Bypassing the nose causes dry gases to reach the respiratory tract and necessitates warming and humidification of gases during mechanical ventilation.

The two nares are divided by a nasal septum that is often not midline. In each naris are three turbinates (Fig. 38-1) that help to condition the inspired gases. Underneath each turbinate lies the opening of perinasal sinus. When these openings are occluded by nasogastric or nasotracheal tubes, fluid tends to accumulate in the sinuses as reflected by a high incidence of radiographic sinus opacification.¹

Figure 38-1



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The nasal turbinates. Note that these are easily traumatized during nasal instrumentation. (Used, with permission, from Pierson DJ, Kacmarek R, eds. *Foundations of Respiratory Care*. New York, NY: Churchill Livingstone; 1992.)

The floor of the nose leading to the nasopharynx is in the same plane as the nasal orifices. When a tube is introduced into the nose, it should be directed straight back rather than caudad (Fig. 38-2), and advanced carefully to avoid injuring the turbinates.

Figure 38-2



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Insertion of a catheter in the nose. The catheter should be directed in parallel with the floor of the nose. (Used, with permission, from Pierson DJ, Kacmarek R, eds. *Foundations of Respiratory Care*. New York, NY: Churchill Livingstone; 1992.)

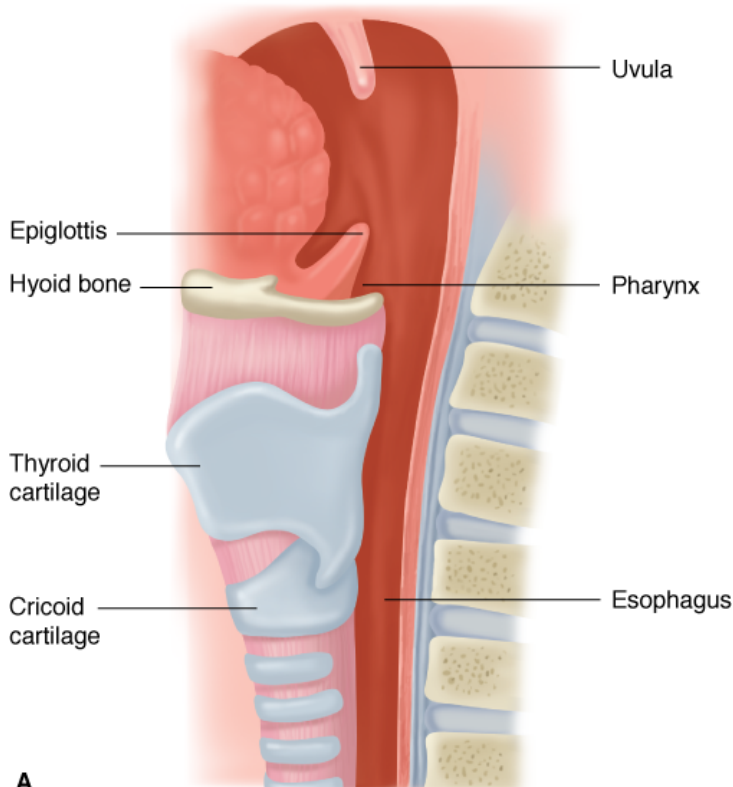
Mouth

Oral structures relevant to airway management include the lips, teeth, and tongue as each may either impede introduction of airway devices into the pharynx or diminish upper airway patency or both during artificial breathing.

Pharynx

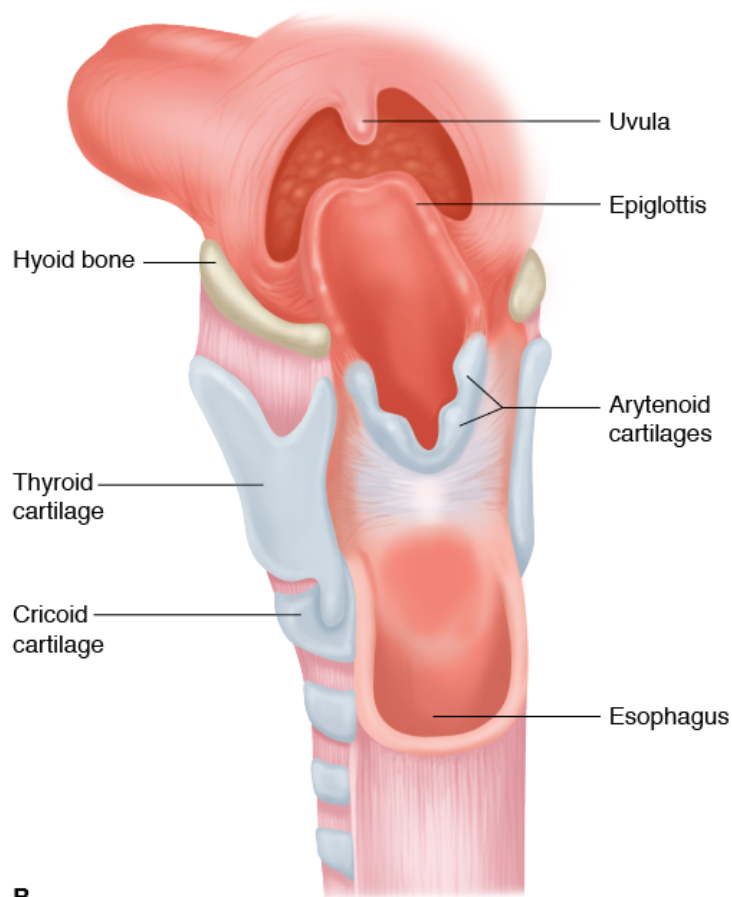
The pharynx is shaped like a cone (Fig. 38-3) and includes the nasopharynx and the oropharynx, which join to form the hypopharynx. The walls of the pharynx are typically soft and compliant, but may become stiff, increase in volume, or both when inflamed from any cause.

Figure 38-3



A

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**B**

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Lateral (A) and posterior oblique (B) views of the pharynx and larynx, including the laryngeal skeleton. (Reproduced, with permission, from Pierson DJ, Kacmarek R, eds. *Foundations of Respiratory Care*. New York, NY: Churchill Livingstone; 1992.)

The digestive and respiratory tracts share a common lumen in the pharynx. The posterior portion of the pharynx continues to form the esophagus whereas the anterior portion ends in a series of pouches or fossae surrounding the larynx. The epiglottis forms the posterior wall of the anterior pouch. The region anterior to the epiglottis is called the vallecula and is an important landmark for endotracheal intubation as the tip of a curved laryngoscope blade is typically placed in this fossa.

The epiglottic cartilage is shaped like a leaf and is attached to the posterior surface of the thyroid cartilage. Functionally, the epiglottis directs food away from the larynx by sitting like a tent over the epiglottic opening. This represents an impediment to visualization of the laryngeal aperture and must be lifted out of the way, either by traction or direct lifting from underneath.

Larynx

The laryngeal skeleton consists of a group of cartilages extending between the fourth and sixth cervical vertebrae. The major portion of the laryngeal body consists of the thyroid cartilage anteriorly and the cricoid cartilage posteriorly. The cricoid cartilage is actually a complete ring with a wide posterior portion and a narrow anterior portion, which is palpable just inferior to the thyroid cartilage. Between the two cartilages is a small depression that represents the cricothyroid membrane.

The arytenoid cartilages sit posteriorly atop the cricoid cartilage. These small paired cartilages provide support to the posterior portions of the true vocal cords. The mucosa of the vocal cords covers these cartilages, and movement of the cartilages occurs during respiration. The arytenoids are an important landmark insofar as they may be the only portion of the glottic inlet visible during attempted laryngoscopy.

Airway Management Without Intubation

Face-mask ventilation, commonly referred to as bag-valve-mask ventilation, is an essential airway management skill. Although considered a “basic” skill, it is not always easy and requires considerable practice, best mastered in a controlled setting such as the operating room. Face-mask ventilation may be required as an initial resuscitative measure, when the need for ventilation is very brief, or when tracheal intubation proves difficult or impossible. Face-mask ventilation may be difficult or impossible because of imperfect patient–face-mask interface or airways obstruction, chiefly of the upper airways but

also of the lower airways. If there is high risk of regurgitation of gastric contents (“full stomach”), active vomiting, or bleeding that might result in massive aspiration, face-mask ventilation should only be used as a rescue technique when severe hypoxemia is present.

Risk Factors for Difficult and Impossible Mask Ventilation

Currently, there is no accepted, single definition of what constitutes difficult mask ventilation. Lack of clinical signs of gas exchange, such as “adequate chest rise” or an [oxygen](#) saturation by pulse oximetry less than 90%, and the need for oral or nasal airways or the use of a second operator to help with face-mask ventilation are the most referenced surrogates for difficulty.² Using these definitions, difficult mask ventilation has been reported in less than 2% of patients.³ Impossible mask ventilation is a rare event, reported in only 1.5/1000 patients.⁴ Caution is advised, however, as these data come from the operating room environment, chiefly in the setting of elective surgery, in patients with stable, managed, or anticipated comorbidities who have been well preoxygenated, rather than in the emergency setting. Risk factors may be segregated by considering operator-dependent factors and patient-related factors, which themselves can be split into those that inhibit proper mask fit and those that may result in upper or lower airways obstruction. [Table 38-1](#) summarizes reported risk factors for difficult and impossible mask ventilation; these factors are highly predictive of difficult mask ventilation, in contrast to the poor positive and negative predictive value of external exams to predict difficult laryngoscopy.^{3–6} When a patient exhibits multiple risk factors for difficult mask ventilation, immediate tracheal intubation (“rapid-sequence”) or an alternative to facemask ventilation, such as placement of a supraglottic airway, should be performed, preferentially at the outset of airways management.

Table 38-1: Risk Factors for Difficult or Impossible Mask Ventilation

Difficult Mask Ventilation
Increased BMI
History of snoring/OSA
Presence of a beard
Lack of teeth
Age older than 55 years
Mallampati III or IV
Limited mandibular protrusion
Male gender
Cervical spine limitation
Airway masses/tumors
Impossible Mask Ventilation
Neck radiation changes
Male gender
OSA
Mallampati III or IV
Presence of a beard

Abbreviations: BMI, body mass index; OSA, obstructive sleep apnea.

Data from Kheterpal et al,^{3,4} Langeron et al,⁵ and Yildiz et al.⁶

Most modern face masks are teardrop shaped, made of clear plastic with large, highly compliant, inflatable borders that conform to a wide variety of faces. A tight seal between the mask and the patient’s face is critical to achieve adequate ventilation. This is most commonly attempted using a generic left-handed technique in which the operator grips with the fifth finger at the left mandibular angle and the third and fourth fingers on the left mandibular ramus while controlling the mask with their first finger and thumb ([Fig. 38-4](#)). In the event that initial face-mask ventilation is difficult or impossible, knowledge of the common anatomic locations of upper airway obstruction allows the airway manager to rapidly perform a variety of maneuvers to relieve the obstruction specific to the site.

Figure 38-4



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The generic, single, left-handed mask-hold technique. The operator grips with the fifth finger at the left mandibular angle and the third and fourth fingers on the left mandibular ramus (in the shape of a letter E) while controlling the mask with his or her first finger and thumb (in the shape of a letter C). The *dotted line* represents the line of the jaw. (Image courtesy A. Joffe.)

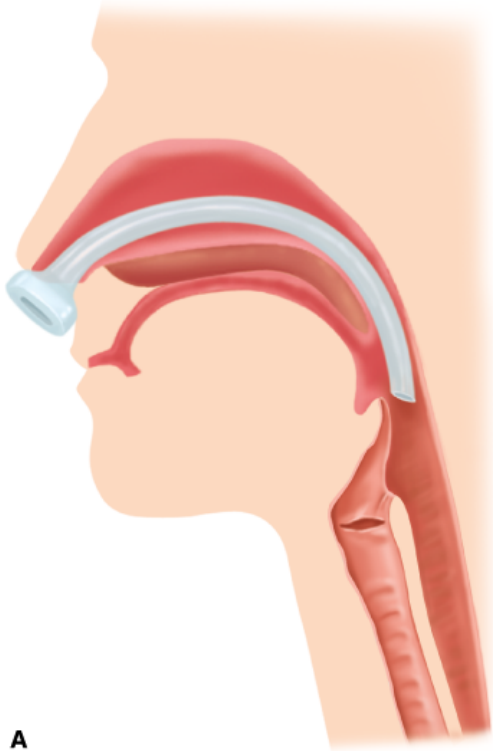
Head Position

As consciousness is lost, upper airway caliber diminishes and obstruction may occur. The so-called sniffing position, used to optimize visualization during direct laryngoscopy, is ideal for providing face-mask ventilation because it decreases the passive collapsibility of the pharynx and pulls the epiglottis and tongue away from the posterior pharynx.⁷ Another maneuver that may open the glottis is to advance the mandible so that the bottom teeth are anterior to the upper teeth.

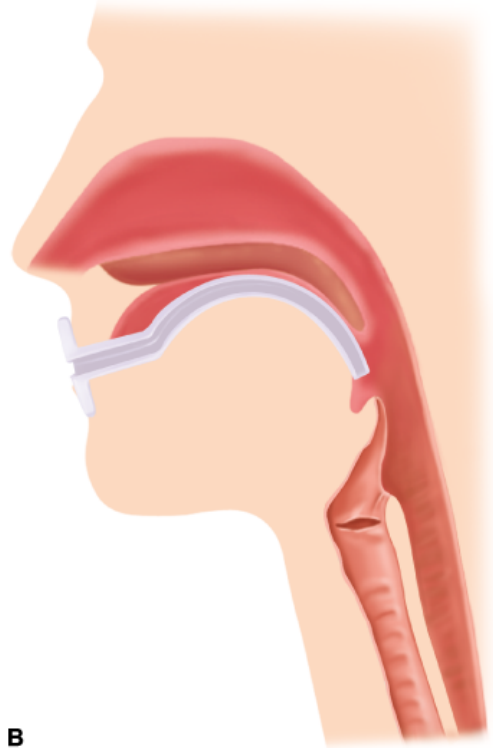
Nasopharyngeal and Oral Airways

The insertion of an oropharyngeal airway maintains the mouth in an open position and bypasses the nasopharynx and base of the tongue. Hypopharyngeal patency may also be augmented. Nasopharyngeal airways relieve obstruction at the level of the nares and soft palate with the potential to improve the patency of the airway at the level of the base of the tongue and hypopharynx (Fig. 38-5). Oropharyngeal airways are tolerated by anesthetized patients, whereas a nasopharyngeal airway is better tolerated in awake individuals. The major hazards of inserting a nasopharyngeal airway are bleeding or fracture of a turbinate. In nonemergent situations, prior application of a topical vasoconstrictor will ease insertion and prevent bleeding. Thrombocytopenia or a coagulation disorder are relative contraindications to insertion of a nasal airway. Proper sizing and careful placement of all pharyngeal airways is important. Improperly placed oropharyngeal airways can push the base of the tongue down into the hypopharynx, while both oropharyngeal and nasopharyngeal airways that are too large for the patient's airway have the potential to displace the epiglottis caudad toward the vocal cord inlet, causing worsened upper airway obstruction.

Figure 38-5

**A**

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**B**

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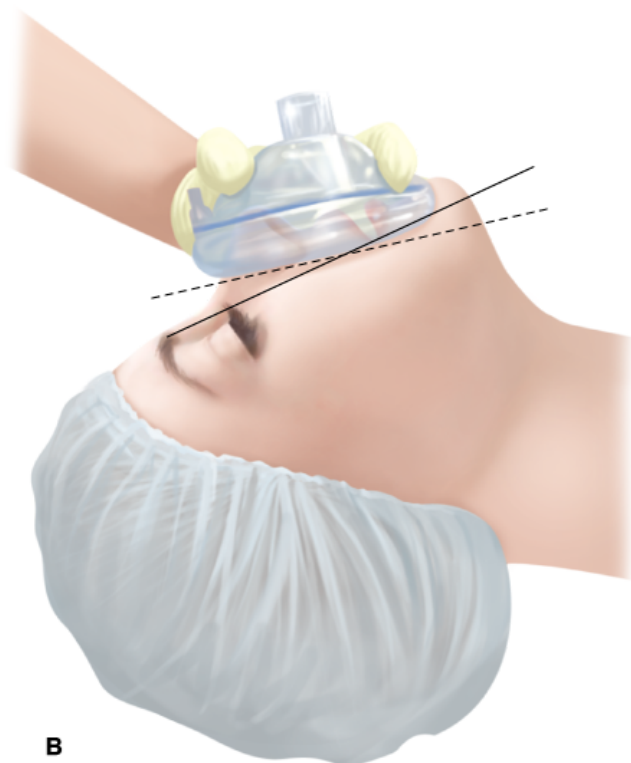
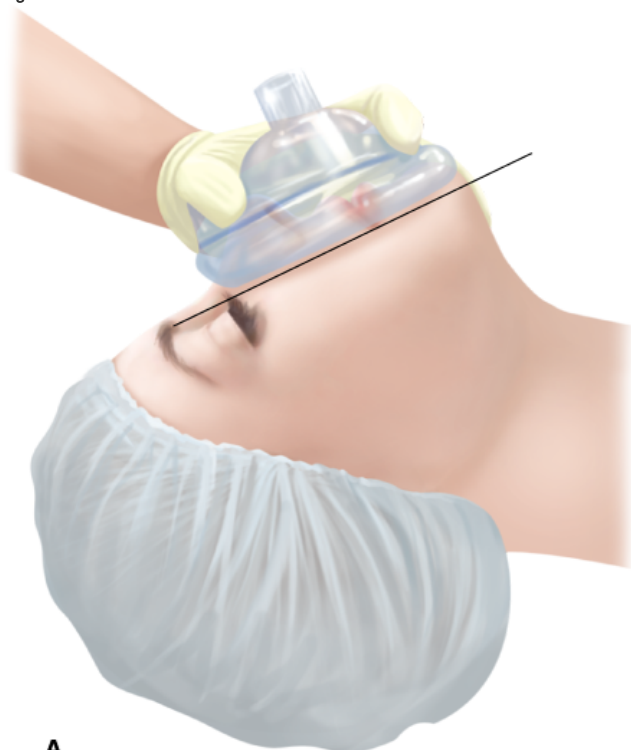
Nasal (A) and oral (B) airways in place. These improve the airway by separating the tongue and epiglottis from the posterior pharyngeal wall. (Used, with permission, from Pierson DJ, Kacmarek R, eds. *Foundations of Respiratory Care*. New York, NY: Churchill Livingstone; 1992.)

Optimizing Face-Mask Ventilation

If initial maneuvers prove unsuccessful in allowing adequate face-mask ventilation, additional techniques include placing the base of the face-mask just caudad to the lower lip rather than into the mandibular groove in edentulous patients; this decreases the leak around the mask and may substantially increase tidal volume (Fig. 38-6).⁸ As Figure 38-7 shows, a two-handed face mask technique with one person securing the mask and a second operator

(out of view in the figure) providing ventilation by squeezing the bag may also increase tidal volumes.⁹ A solo operator can also utilize a two-handed mask technique by using an intensive care unit ventilator set to deliver pressure-control ventilation with a limit of 20 cm H₂O as the “second operator.”¹⁰ In patients with factors that predict difficult mask ventilation, it may be advisable to employ all available maneuvers initially to provide for an optimal first attempt at face-mask ventilation.

Figure 38-6



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A comparison of the traditional location of the mask in relation to the patient's face (*top*) with the top of the mask over the bridge of the nose and the base of the mask over the mandibular groove. The *continuous line* represents the plane between the nose and the mandible. When the mask is moved

caudally (*bottom*), the base of the mask becomes positioned above the lower lip. The *dashed line* represents the contact between the mask and the face. This “lower lip” placement is particularly advantageous in edentulous patients. (Images courtesy A. Joffe.)

Figure 38-7



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A two-handed jaw-thrust mask-hold technique. The key to the maneuver is providing sufficient anterior traction to effect maximal mandibular advancement such that the mandibular teeth are forward of the maxillary teeth. This pulls the tissues supported by the mandible—the tongue and epiglottis—away from the posterior pharynx, thereby increasing the airway caliber. In addition, traction may increase the patency of the glottic aperture. (Photograph courtesy A. Joffe.)

Tracheal Intubation: Indications, Preparation, and Standard Techniques

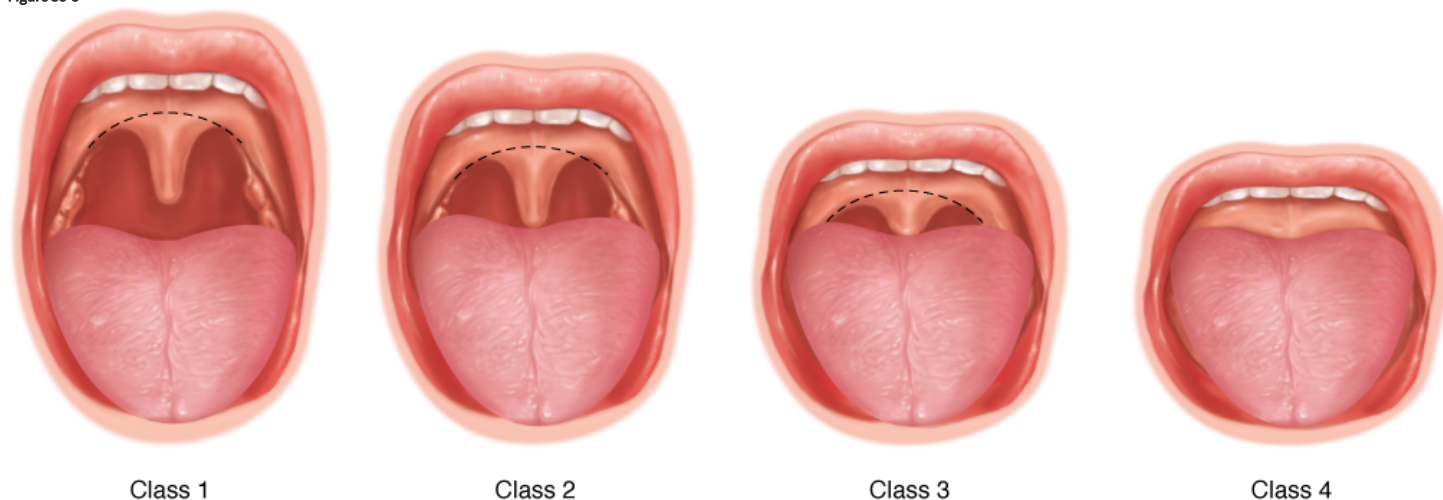
The indications for tracheal intubation are varied, but for the most part can be categorized into (a) the need for maintenance and protection of the airway, (b) the need for high inspired [oxygen](#) concentrations, or (c) the need for application of positive pressure to the airway. Significant overlap may exist in individual patients. Specific etiologies of respiratory failure and the need for mechanical ventilation are covered in detail in [Chapter 4](#).

Airway Evaluation

A substantial body of research has been directed toward identification of patients with “difficult airways”; unfortunately, anatomy-based predictors of the difficult airway are notoriously insensitive and have only moderate specificity.¹¹⁻¹³ Nonetheless, a basic history should be obtained and an airway examination performed to the extent possible in all patients to identify risk factors for difficult ventilation and/or intubation. Factors associated with difficult intubation include a previous history difficult intubation; morbid obesity;^{14,15} obstructive sleep apnea;^{16,17} and gross physical abnormalities, including marked prognathia or retrognathia, other congenital deformities, a marked overbite, facial swelling, postradiation fibrosis, and facial dressings. A basic examination includes a view of the patient’s facial anatomy from the frontal and lateral perspectives, an oral examination with the mouth maximally open and tongue extended, an assessment of neck extension, and measurement of the distance from the tip of the mandible to the thyroid cartilage. Additional factors to consider in evaluating the airway are discussed below.

Based on oral examination, Mallampati devised a classification scheme to predict difficult intubation ([Fig. 38-8](#)).¹⁸ In general, the likelihood of difficult intubation increases from Class 1 to Class 4, although the test has only approximately 50% sensitivity at identifying difficult airways, and is most useful as a positive predictor at the extreme (Class 4).^{18,19} In addition, this test is often impractical in emergency settings, where patient cooperation with the examination is not possible.

Figure 38-8



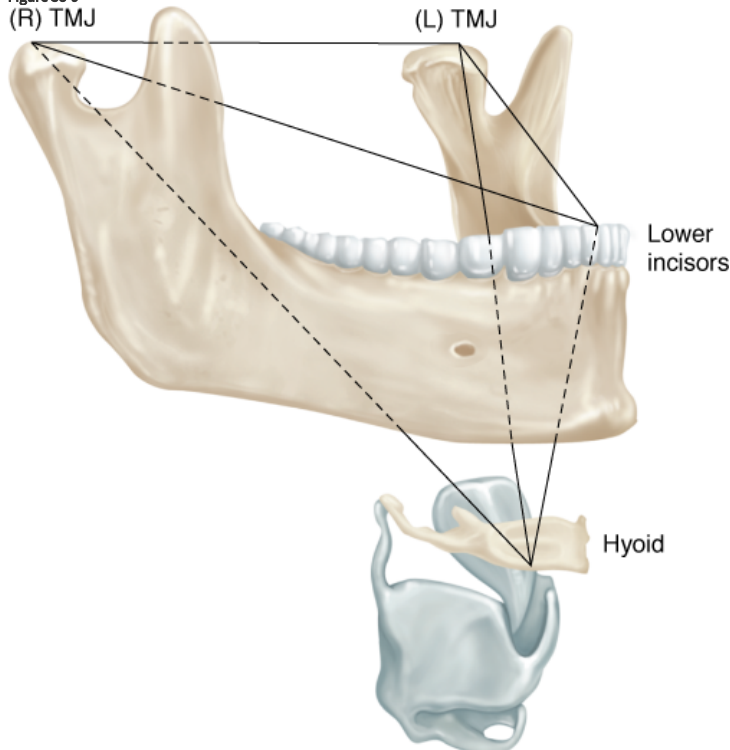
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Mallampati Airway Classification. *From left to right:* Class 1: soft palate, fauces, uvula, and anterior and posterior tonsillar pillars visible; Class 2: soft palate, fauces, uvula visible; Class 3: soft palate, base of uvula visible; and Class 4: hard palate only.

The size of the mandible provides some measure of the space available to displace the tongue during laryngoscopy (Fig. 38-9). This can be assessed by measurement of the distance between the tip of the mandible and the thyroid cartilage (thyromental distance) using either a ruler or finger-breadths that have previously been calibrated against a ruler. A thyromental distance of less than 6 to 7 cm has a specificity of approximately 90% in identifying the difficult airway, but the test has very poor sensitivity (approximately 30%).¹² Complementary to the absolute distance between the mentum and thyroid cartilage is the potential distance as assessed by the upper-lip bite test. To perform the test, the patient is simply asked to bite his or her upper lip and is classified as follows: Class 1, lower incisors can bite the upper lip above the vermilion line; Class 2, lower incisors can bite the upper lip below the vermilion line; and Class 3, lower incisors cannot bite the upper lip (Fig. 38-10). Specificity and accuracy of the upper-lip bite test are significantly higher than that reported for thyromental distance alone.²⁰

Figure 38-9



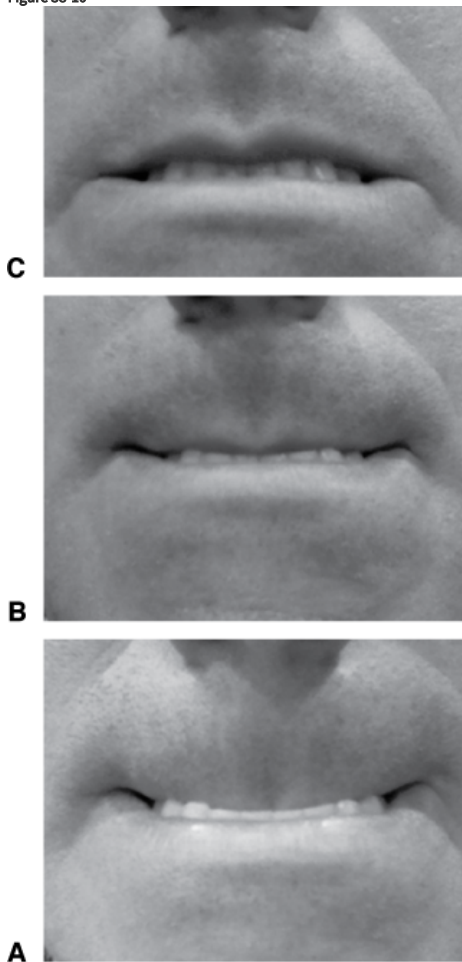
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The submandibular space represented by an inverted pyramid bounded by the right and left temporomandibular joints (TMJs), lower incisors, and hyoid bone. This represents the potential space for tissues to be displaced anteriorly and away from the posterior pharynx and lines of sight during airway-

management maneuvers. (Adapted, with permission, from Greenland KB. A proposed model for direct laryngoscopy and tracheal intubation. *Anaesthesia*. 2008;63:156–161.)

Figure 38-10



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A frontal view of the upper-lip bite test. *From bottom to top (best to worst)*: In Class 1, the lower incisors are able to bite the upper lip past the vermilion border, making the mucosa of the upper lip totally invisible (**A**). In Class 2, a portion of the lip below the vermilion border remains visible (**B**). In Class 3, the lower incisors fail to bite any portion of the upper lip (**C**). (Images courtesy A. Joffe.)

Extension of the head on the neck is necessary to bring the pharyngeal and laryngeal axes into alignment during direct laryngoscopy. Head and neck mobility can be assessed by simple examination, or quantitated by measuring the distance from the sternum to the tip of the mandible (sternomental distance) at maximum neck extension.²¹ A sternomental distance of equal to or less than 13.5 cm has greater than 50% sensitivity and specificity at predicting difficult intubation, although this test has not been studied as extensively as either the Mallampati classification or the thyromental distance.²⁰

Common sense dictates that a combination of the above tests may prove more useful at predicting the difficult airway than the individual tests. There is some literature to support this notion, although the predictive value of these tests remains modest even when used in combination.^{22,23} When possible in the context of the critically ill patient, a minimal exam, which assesses mouth opening, the size of the tongue, cervical spine motion, and mandibular protrusion may be helpful to assess potential problems with laryngoscopy.

Other factors that may complicate emergency intubation and require special consideration include:

1. Risk factors for regurgitation and tracheal aspiration of gastric contents, including recent oral intake; gastroparesis because of drugs; intraabdominal infection, trauma, or diabetic neuropathy; increased intraabdominal pressure (pregnancy, ascites); and bowel obstruction. Most critically ill patients should be considered at risk for regurgitation and aspiration during intubation.
2. Hypovolemia and/or shock predisposes to severe hemodynamic depression during tracheal intubation because of sedative/hypnotic drugs and the initiation of positive pressure ventilation. Vascular access, drugs, fluids, and equipment for resuscitation should be present.

3. Increased intracranial pressure, which can be exacerbated by direct laryngoscopy and tracheal intubation.
4. Acute trauma poses multiple problems during attempted intubation, including possible airway injury, head and brain injury, cervical spine injury, and hypovolemia. Intubation in the presence of known or possible cervical spine injury must proceed without movement of the spine (inline manual spine stabilization).
5. Severe hypoxemia reduces the time available for laryngoscopy and intubation before severe oxyhemoglobin desaturation occurs.

Equipment and Setup

Table 38-2 lists supplies that should be present at the site of tracheal intubation, some of which may be institution specific, particularly the equipment available for the anticipated or unanticipated difficult airway. In emergent situations, the endotracheal tube (ETT) should be prefitted with a stylet with a slight anterior bend on the tip (“hockey stick”), which may facilitate placement if the laryngeal view is suboptimal.

Table 38-2: Supplies for Adult Tracheal Intubation

Routine Supplies
Standard disposable inflatable-rimmed teardrop-shaped face mask in small, regular, and large adult sizes Oropharyngeal (80 to 100 mm) and nasopharyngeal (28 to 32 Fr) airways Self-inflating resuscitator circuit (“Ambu Bag” or modified Mapleson circuit) Yankauer-type suction catheter Standard and short laryngoscope handle Blade of choice (curved or straight) in two sizes Tracheal tube introducer (Eschmann, Frova, etc.) Supraglottic airways to accommodate small to large adults (30 to 100 kg, laryngeal mask, laryngeal tube, other) Tracheal tubes (size 7- to 8-mm internal diameter) Resuscitation drugs and defibrillator immediately available Device for tube placement confirmation (carbon dioxide detector, self-inflating bulb, other)
Medications
Sedative-hypnotics Neuromuscular blocking drugs
Additional Difficult Airway Equipment
Indirect video laryngoscope (GlideScope, Airtraq, other) Intubating laryngeal mask airway (LMA-Fastrach, Air-Q) Fiber-optic bronchoscope Aintree intubating catheter, Arndt wire-guided catheter exchange set Prepackaged cricothyrotomy kit (Melker, Arndt, other) <i>or</i> #11 or 20 scalpel with 6.0 inner diameter tracheal tube for cricothyrotomy

Preparation for emergency tracheal intubation by following a checklist or bundle is recommended. Use of an intubation bundle incorporating preoxygenation, the presence of two operators, a rapid sequence induction, cricoid pressure, postintubation capnography, low-tidal volume lung-protective ventilation, fluid loading, and preparation and early administration of sedative and vasopressor infusions has been reported to significantly reduce the occurrences of cardiac arrest or death, severe cardiovascular collapse, and hypoxemia occurring within 60 minutes of emergency intubation in the intensive care unit (Table 38-3).²⁴

Table 38-3: Intubation Care Bundle

Preintubation
1. Presence of two operators 2. Fluid loading (isotonic saline 500 mL or equivalent) in absence of cardiogenic pulmonary edema 3. Preparation of intensive care unit sedatives (bolus or infusions) 4. Preoxygenate (3 to 5 minutes with tight-fitting high-flow non-rebreather mask or noninvasive positive pressure ventilation in case of acute respiratory failure with FI_{O_2} 100%, pressure support 5 to 15 cm H₂O to obtain expiratory tidal volume of 6 to 8 mL/kg and PEEP of 5 cm H₂O)
Intubation
5. Rapid sequence induction, with or without application of cricoid pressure (see Table 38-4)
Postintubation
6. Immediate confirmation of tube placement 7. Vasopressor administration for postintubation hypotension 8. Initiate sedation 9. Initial ventilator settings to achieve lung protection

Abbreviations: FI_{O_2} , fractional inspired oxygen concentration; PEEP, positive end-expiratory pressure.

Adapted, with permission, from Jaber et al.²⁴

Standard Intubation Techniques

Oral versus Nasal Intubation

The decision to place an oral or nasal tracheal tube is based on a variety of considerations. In general, the oral route is preferred. A larger tube can be placed orally than nasally, resulting in less airways resistance,²⁵ higher minute ventilation, easier passage of fiber-optic bronchoscopes, and more effective pulmonary toilet. Nasal tubes occlude the sinus ostia, potentially leading to a higher incidence of bacterial sinusitis compared to oral tubes.^{1,26} Nasal intubation is contraindicated in the presence of coagulopathy or anticoagulation, basilar skull fracture, or significant nasal or sinus deformity.

Patient Preparation: Rapid Sequence Induction of Anesthesia

In most cases, the most rapid way to secure the airway is to induce general anesthesia with a sedative and/or hypnotic, followed by administration of a neuromuscular blocking drug, followed by direct laryngoscopy with tracheal intubation. Table 38-4 describes a “rapid sequence induction.” This approach, when combined with the application of cricoid pressure (Sellick maneuver) (Fig. 38-11), may minimize the chances of massive aspiration. Note that recent investigations have determined that the mechanism by which cricoid pressure may prevent regurgitation of gastric contents into the pharynx is related to occlusion of the hypopharynx (rather than the esophagus) by downward pressure on the cricoid cartilage.^{27,28}

Table 38-4: Steps for Rapid-Sequence Induction and Tracheal Intubation

1. Prepare equipment and position patient; use of styletted tubes is recommended
2. Preoxygenate patient (this may be occurring simultaneously or can precede #1 above)
3. Apply cricoid pressure (optional depending on circumstance)
4. Rapid administration of induction agent followed by a neuromuscular blocking drug
5. Wait for fasciculations (succinylcholine only) or a full 60 seconds (either succinylcholine or rocuronium), whichever comes first^a
6. Direct laryngoscopy and tracheal intubation^b
7. Verification of correct tube placement

^aDepending on circumstances (suspected full stomach or not), mask ventilation may be briefly attempted before intubation attempts. The ability to provide mask ventilation assures the operator that if laryngoscopy is difficult or impossible, face-mask ventilation will suffice while other intubation pathways are pursued. If face-mask ventilation is unsuccessful in this circumstance, immediate placement of a supraglottic airway or performance of cricothyroidotomy may be necessary should initial intubation attempts fail.

^bInitial attempts may be made by indirect video laryngoscopy via a supraglottic airway, or by fiber-optic bronchoscope, depending on the situation and the experience of the operator.

Figure 38-11

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(A) A sagittal cadaver image of the cricoid pressure anatomic unit. The cricoid cartilage (C) is the foundation of the cricoid-pressure unit with its superior and inferior aspects indicated by *black arrows*. The postcricoid hypopharynx muscular wall and lumen (*white arrowhead*) and its superior and inferior limits (*white arrows*) have a constant and intimate relationship with the cricoid cartilage. As pressure is applied anteriorly (*serial black arrows*), the unit is compressed against the vertebral body (C5), sealing off the esophagus (E) at the esophageal verge (EV). Because the esophagus is well below the level of the seal, its position is irrelevant to the integrity of the seal produced by the pressure on the unit. (B) An image from Sellick's original report showing obliteration of the lumen by cricoid pressure at the fifth cervical vertebra. T, trachea. (Used, with permission, from Rice MJ, Mancuso AA, Gibbs C, et al. Cricoid pressure results in compression of the postcricoid hypopharynx: the esophageal position is irrelevant. *Anesth Analg*. 2009;109(5):1546–1552.)

“Preoxygenation,” or “denitrogenation,” consists of allowing the patient to breathe 100% oxygen for 3 to 5 minutes before induction of anesthesia. Preoxygenation allows utilization of the functional residual capacity as an oxygen reservoir. When functional residual capacity and oxygen consumption are normal, preoxygenation allows 8 or more minutes of apnea before oxyhemoglobin desaturation, and greatly increases the margin of safety should there be difficulties in managing the airway.²⁹ Preoxygenation, however, is far less effective in patients with acute hypoxemic respiratory failure,³⁰ and increasing the duration of preoxygenation from 4 to 8 minutes is of only marginal benefit in these patients.³¹ On the other hand, the use of noninvasive positive-pressure ventilation in critically ill, hypoxemic patients provides more thorough denitrogenation and longer apnea times than tidal breathing with a non–rebreather mask, and should be considered when staff and equipment are readily available.³² At a minimum, if patients are already receiving noninvasive positive-pressure ventilation before intubation, it should be continued until laryngoscopy rather than switching to conventional face-mask preoxygenation.

If rapid sequence induction of anesthesia is not indicated, induction may proceed at a more measured pace, without application of cricoid pressure. On the other hand, preoxygenation should be attempted in all emergency intubations, recognizing that it may be neither possible nor effective in some circumstances.

Direct Laryngoscopy

Intubation using direct laryngoscopy may be the fastest and surest way of securing an airway. This may be particularly true where use of video laryngoscopes or a fiber-optic bronchoscope may be hindered by airway soiling with secretions, blood, or vomitus. This important lifesaving skill, however, can result in great harm if performed incorrectly. The major, feared complications are failure to ventilate and intubate the trachea, and unrecognized esophageal intubation, both of which can lead to severe neurologic injury and death. The likelihood of successful intubation is increased with experience, in addition to the use of proper equipment and suitable pharmacologic adjuncts. Gaining expertise in manual face-mask ventilation and direct laryngoscopy in a controlled setting before attempting these maneuvers in an emergency cannot be overemphasized. The specifics and technical considerations of direct laryngoscopy are beyond the scope of this chapter. A brief description of the basic technique follows.

Numerous laryngoscope blades have been devised for various special situations but most intubations are performed using one of two categories of blades: (a) blades intended to be used to lift the epiglottis directly (“straight” blades); (b) blades designed to lift the epiglottis indirectly by placing anterior traction on the hyoepiglottic ligament in the vallecula (“curved” blades) (Fig. 38-12). In our experience, the curved laryngoscope blade is more easily mastered and more commonly used than the straight blade, although facility with both is ideal as each has distinct advantages. Laryngoscope blades are most commonly designed to be held by the left hand (for right-handed individuals), although right-handed blades are available.

Figure 38-12



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Typical straight (“Miller,” *top*) and curved (“Macintosh,” *below*) blades for direct laryngoscopy. Straight blades directly lift the epiglottis. Although this often provides a clear view of the glottis, the room available for inserting the endotracheal tube into the mouth is limited compared to curved blade. The curved blade, the most common variant of which is the Macintosh, is directed into the vallecula and then pulled forward. This lifts the epiglottis away from the glottis. (Photograph courtesy S. Deem.)

Before starting, the patient’s head should be positioned near the top of the bed so the laryngoscopist’s arms are not fully extended. The force applied during the average intubation is 25 newtons (N),³³ the equivalent of lifting a mass of approximately 2.5 kg. This force is most easily applied when the elbow is flexed and relatively close to the laryngoscopist’s body.

Classical teaching is that laryngoscopy is facilitated by flexion of the neck at the cervical–thoracic junction and extension of the neck at the atlantooccipital joint (“sniffing position”); this position brings the laryngeal and pharyngeal axes into alignment. The sniffing position is accomplished by raising the occiput with a firm pillow, folded towels, or other support. Recent studies, however, suggest that cervical extension is the most important component of this maneuver, and that the addition of neck flexion on the thorax improves the laryngoscopic view only in patients with limited neck extension and/or morbid obesity.^{34,35}

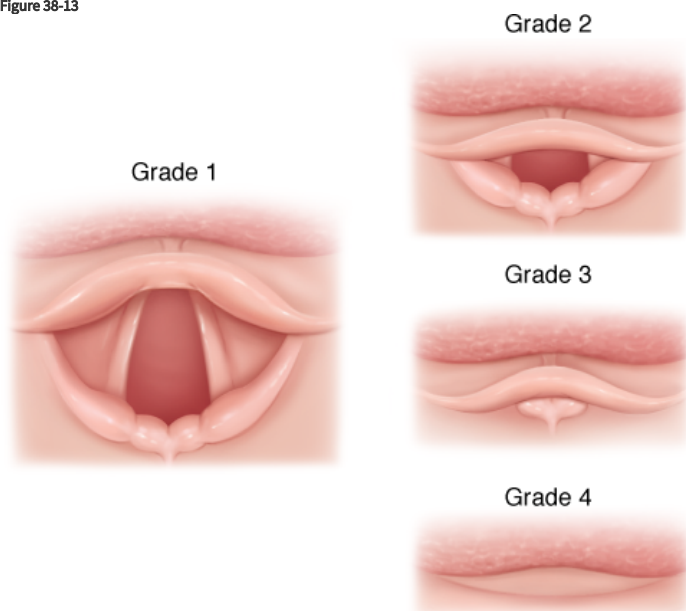
Laryngoscopy proper begins by positioning the blade in the right side of the patient’s mouth; the mouth opening can be facilitated with the right hand. The right hand can then move to the occiput to extend or flex the neck into optimal position. The blade is then slowly advanced and the tongue swept to the left. The uvula and then the epiglottis should be identified, and are useful landmarks for maintaining a midline position. The blade should be advanced either into the vallecula (curved blade), or beyond the epiglottis toward the glottis (straight blade), and traction then applied at a 45-degree angle with floor. The wrist should remain firm, as flexing the wrist may cause pressure on the upper incisors and will push the epiglottis in front of the larynx. Traction at a 45-degree angle pulls the epiglottis forward to reveal the glottic opening. If the glottis is still not visible, external application of backward (toward the spine), upward (toward the head), rightward (toward the patient’s right side) pressure on the thyroid cartilage (“BURP” maneuver) may push the larynx into view; an assistant can then hold the larynx in this position.³⁶ Once the laryngeal aperture is visualized, the ETT can be advanced through the glottis, the stylet removed, and the cuff inflated; the correct position should be verified as described below.

If the nasal route is used, application of a topical vasoconstrictor before intubation is of critical importance in increasing the size of the nasal passage and minimizing bleeding. Topical 4% cocaine is effective, providing both vasoconstriction and anesthesia, but may be toxic at doses greater than 3 to 4 mg/kg. An effective alternative is to use a [phenylephrine](#) or [oxymetazoline](#) nasal spray several minutes before instrumenting the nose, particularly if topical anesthesia is not necessary. Both nostrils should be sprayed in case one side is partially obstructed. The nasal passage can be prelubricated by inserting a nasopharyngeal airway coated with either 2% [lidocaine](#) gel (for awake intubations) or water-soluble gel. In addition, briefly warming the ETT in 37°C (98.6°F) water will make it much more pliable and less likely to injure the mucosa. Nasal tubes (without a stylet) should be inserted with gentle pressure, directed posteriorly and advanced into the pharynx. Direct laryngoscopy is then performed, and the tube is guided into the larynx by manipulating the end extending out of the nose. Frequently, the tube tip must be guided with a Magill forceps while an assistant advances the tube.

The Difficult Airway

The term *difficult airway* is best defined as the development of any airway management problem that requires an escalation of interventions to establish mask ventilation and/or tracheal intubation. For purposes of standardization in studies, difficult intubation is often defined as that requiring more than two attempts at laryngoscopy, or by a grade 3 or 4 laryngoscopic view as described by Cormack and Lehane (Fig. 38-13), although considerable subjectivity exists for all definitions.³⁷ “Failed” intubation implies that either attempts were aborted, or that the airway was secured surgically. The estimated incidence of difficult tracheal intubation in the operating room is approximately 5%, and failed intubation is less than 0.5%.¹² The reported incidence of difficult intubation in the emergency setting varies widely, but on the average is similar to that for the operating room. The reported incidence of failed intubation, however, is much higher (up to 5%), as is the incidence of other intubation-associated complications such as hypoxemia, aspiration, cardiac arrest, and death.^{38–41}

Figure 38-13



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Laryngoscopy grading, usually referred to as Cormack-Lehane grade after the physicians who originally described them. *From left to right:* Grade 1: entire glottic opening can be visualized; Grade 2: only the posterior glottis or posterior arytenoids can be visualized; Grade 3: only the epiglottis is visible; Grade 4: no recognizable structures are visible.

It is not surprising that difficulties in managing the airway are more likely to occur during emergency intubation than during an elective situation given the comorbidities that may complicate emergency intubation. Emergency intubation also poses an additional problem in that the patient cannot be awakened and the intubation postponed, particularly as critically ill patients often require intubation because of inadequate oxygenation and/or ventilation.

Guidelines for management of the difficult airway based on available literature and expert opinion have been developed locally (at the hospital or practice level) and regionally or nationally.^{12,42–45} The American Society of Anesthesiology was the first organization to publish such guidelines (including an accompanying algorithm), with the last update several years old at the time of this writing.⁴³ It is the general perception that difficult airway guidelines and algorithms have resulted in fewer airway mishaps. This appears to be true regarding airway management at the induction of anesthesia. The same conclusion, however, is not supported with respect to airway management during other phases of anesthesia or in locations outside of the operating room.⁴⁶ Thus, other groups have developed guidelines that are more prescriptive.⁴⁵

Despite the lack of hard supporting evidence, common sense dictates that all airway providers should gain expertise in difficult airway management skills, and should have an algorithm for dealing with the difficult airway. Which published algorithm iteration is used is likely not as important as the act of learning and implementing it. Institutional specific algorithms based upon the types of patients to be cared for, the local experience, and available equipment may have a greater impact than broader, society-based difficult airway algorithms. Recent reports support this notion for elective surgical patients and in the prehospital setting.^{47,48} Every institution should have an organized team for responding to critical airways, with minimal training standards for all designated responders, equipment standards, and a mechanism for providing the rare emergency surgical airway.

Predicted Difficult Airway: Awake Intubation

If the patient has a recognized difficult airway and is not combative or uncooperative, an awake intubation is generally preferable. This can be accomplished by one of several techniques, including blindly placing a nasotracheal tube, placing an oral or nasal tube using fiber-optic guidance, using

awake direct laryngoscopy, or using a retrograde tube-over-guide technique. Alternatively, the airway may be surgically secured primarily by an awake tracheotomy or, in emergent situations, by an emergency cricothyrotomy. Preparation of the patient and some awake intubation techniques are outlined below.

Patient Preparation for Awake Intubation

Awake intubation can be performed with surprisingly little discomfort to the patient if topical anesthesia is first applied. Administration of an antisialagogue such as [glycopyrrolate](#) decreases secretions that impair visibility and enables better uptake of local anesthetics by the mucous membranes. If intubation is emergent or urgent, there may be insufficient time for the antisialagogue to become effective, however.

Topical anesthesia of the soft palate, pharynx, and larynx can be obtained by administration of atomized or nebulized [lidocaine](#) (2% to 4%) or benzocaine spray, and/or by asking the patient to gargle with viscous [lidocaine](#) solution. Caution is warranted regarding the use of benzocaine, as methemoglobinemia has been reported with this agent, and toxicity is not necessarily related to the administered dose.^{49,50} Patients receiving benzocaine should be monitored for the development of cyanosis; the arterial saturation as estimated by pulse oximetry may fall, but this monitor underestimates the true oxyhemoglobin saturation.⁵¹ A high level of suspicion is necessary, and methemoglobinemia can be confirmed by cooximetry. If [lidocaine](#) is used, the total administered dose should be limited to less than 300 mg as absorption via the mucosa is significant and can lead to central nervous system toxicity.

Anesthesia of the trachea can be achieved by nebulization of [lidocaine](#), by spraying [lidocaine](#) through a fiber-optic bronchoscope, or via transtracheal injection of [lidocaine](#), with the transtracheal approach providing the best anesthesia.⁵² This can be accomplished by rapid injection of 2 to 3 mL of 4% [lidocaine](#) with a 22-gauge needle at end-exhalation; the subsequent cough helps distribute the anesthetic.

Direct or indirect video laryngoscopy can be performed in awake patients, provided that adequate topical anesthesia has been applied, although supplemental sedative may be required. Additional awake techniques are discussed below.

Awake Blind Nasal

Blind nasal intubation has the advantage of not requiring a laryngoscope (or bronchoscope), can be performed even when blood or secretions are present in the airway, and can be performed on a patient in the sitting position. The technique is especially suited to dyspneic patients, as they breathe more comfortably when sitting, have easily heard breath sounds, and tend to maintain their glottis open. Once the tube is in the nasopharynx, it should be directed caudad, using breath sounds as a guide, and advanced through the glottis during inspiration. The course of the tube can sometimes be determined by external palpation of the neck. The larynx can often be pushed gently toward one side if the tube is slightly misaligned. Rotation of the tube and flexion or extension of the neck may also be useful if the tube does not initially enter the trachea. The overall success rate for blind nasal intubation is approximately 90% in experienced hands.⁵³

Fiber-Optic Tracheal Intubation

Fiber-optic-aided intubation allows direct visual guidance, can be accomplished in the awake patient via either the nasal or oral route, and is less stimulating than direct laryngoscopy. In emergency situations, however, when blood, vomitus, or copious secretions are present, it may be quite difficult.

Preparation of the patient for awake fiber-optic intubation is described under Patient Preparation for Awake Intubation; the neck should be neutral or slightly extended. The scope and light source should be verified to be in working order, the bronchoscope lubricated, and the tube passed over it to ensure proper fit. [Oxygen](#) can be insufflated through the suction port of the scope, or direct through the patient's nose or mouth, depending on what intubation route is used. The nasal route may provide the best angle for viewing and entering the larynx. The ETT can be passed into the nose and advanced into the nasopharynx, and the bronchoscope then introduced. If the oral route is chosen, an oral intubating airway may facilitate passage of the scope through the mouth and pharynx. Alternatively, an assistant can grasp and gently pull the patient's tongue with a gauze pad. This opens the pharynx and facilitates visualization of the larynx. If the patient is sedated or uncooperative, a bite block should be inserted to prevent biting of the scope. After the scope enters the trachea, the ETT can be advanced, and its position relative to the carina confirmed before withdrawing the scope. If the tube does not easily pass through the glottis, a 90-degree clockwise rotation may allow passage beyond the right vocal cord.

Video (Indirect) Laryngoscopy

A number of indirect laryngoscopes with video capability are commercially available and are generally classified into steering and nonsteering devices, although it is also useful to differentiate them by whether or not the device incorporates an integrated channel for delivery of the tracheal tube. Five devices, two of which are channeled, are currently available or soon will be. [Table 38-5](#) compares the features unique or shared amongst the devices.

Table 38-5: A Comparison of Some Commonly Used Video Laryngoscopes

Device	Blade Shape	Monitor	Reusable/Disposable	Size Range
GlideScope	Angulated	Separate, 3.5-in. (ranger) or 7-in. LCD monitor	Reusable, single-use blades (cobalt, ranger)	Pediatric–large adult, depending on model
Storz C-Mac, D-Mac	Standard Macintosh blade, angulated	Separate, 7-in. TFT monitor	Reusable	Sizes 2 to 4
McGrath	Angulated	Integrated, 1.7-in. LCD monitor	Single-use blades	Three adult lengths
Pentax-AWS	Anatomically shaped with guide channel	Integrated, 2.4-in. LCD monitor	Single-use blades	One size only
Airtraq	Anatomically shaped with guide channel	None (with eyepiece) or external monitor	Single-use blades	Pediatric, small adult, regular adult, double-lumen

Abbreviations: LCD, liquid crystal display; TFT, thin-film transistor.

Data from Niforopoulou P, Pantazopoulos I, Demestiha T, et al. Video-laryngoscopes in the adult airway management: a topical review of the literature. *Acta Anaesthesiol Scand*. 2010;54(9):1050–1061.¹²⁸

The most important feature shared amongst videolaryngoscopes is their ability to “look around the corner” enabling an almost universally improved view of the glottic structures compared to those obtained with direct laryngoscopy; the technique has recently been reviewed in detail.⁵⁴ Videolaryngoscopes can be used preferentially as an “optimal” first attempt at intubation, for rescue after failed direct laryngoscopy, after topicalization of the airway for awake intubation, or to inspect the airway and rule out airway edema or obstructing masses before extubation of the difficult airway.

The Unrecognized Difficult Airway

The term *unrecognized* implies that the initial attempt by any method of airway instrumentation, but chiefly direct laryngoscopy, was not straightforward as expected. As mentioned previously, prediction of difficult intubation is an inexact science, and it is not uncommon to have unexpected difficulties with tracheal intubation after induction of anesthesia and paralysis. Furthermore, in the critically ill patient, awake intubation or postponement of intubation are frequently not options either because of patient combativeness or the immediate need for control of the airway or severe hypoxemia. Thus, perhaps the most essential component of airway management skills is the ability to rapidly and systematically respond to difficulties with intubation and/or face-mask ventilation. At the first sign of airway difficulties, help should be summoned, difficult airway equipment obtained if not already present, and a difficult airway algorithm followed.

The first laryngoscopy attempt should be the optimal attempt and thus particular attention must be paid to equipment setup, patient positioning, ergonomics of the operator, and the surrounding conditions such as support staff, lighting, and ambient noise. Potential technique modifications occurring at the time of laryngoscopy include adjustment of head and neck position, and external laryngeal manipulation/ bimanual laryngoscopy (“BURP” maneuver). Based upon the best laryngoscopic view obtained, intubation can be attempted by direct placement of the tracheal tube, use of an Eschmann introducer or bougie, or immediate switch to an alternate laryngoscope blade or device for a second attempt. For example, in large patients where the blade is not long enough to reach the vallecula, a longer blade (i.e., Macintosh #4 instead of Macintosh #3 blade) may be useful; in patients with a long, floppy epiglottis, switching from a curved to a straight blade may be helpful. If initial intubation attempts fail but mask ventilation is adequate, the situation remains controlled. In general, however, no more than two attempts at direct laryngoscopy should be made before choosing an alternate technique, as multiple laryngoscopies are associated with increased morbidity.

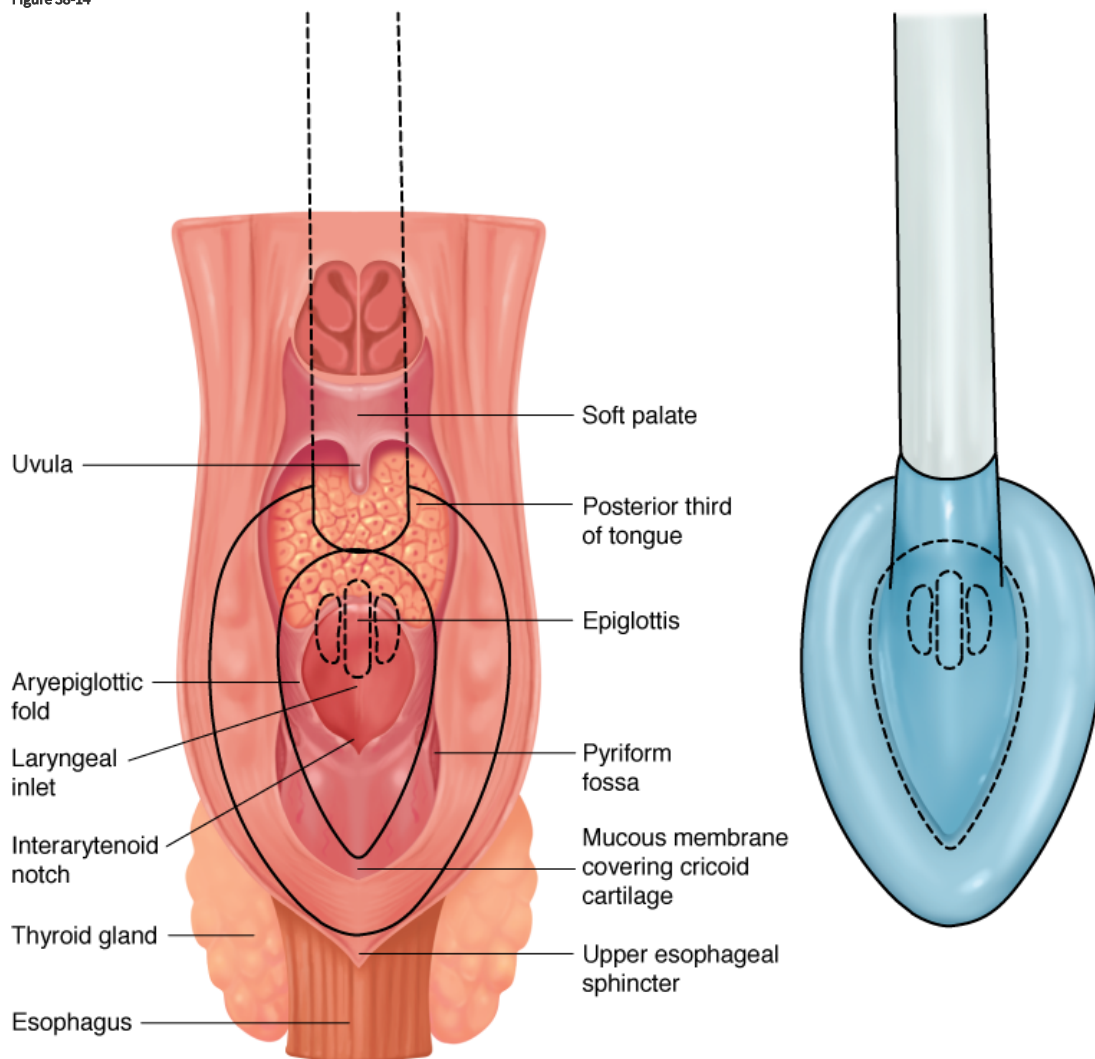
If mask ventilation remains effective, other alternate techniques to consider are video laryngoscopy, fiber-optic intubation, or laryngeal mask-facilitated intubation. If these techniques are not practical for any reason, remaining options are limited, and include a surgical airway and awakening the patient. Awakening the patient, however, is infrequently an option in the emergency setting, as discussed above.

Difficult Intubation and Difficult Face-Mask Ventilation

If initial intubation attempts fail and face-mask ventilation becomes inadequate, a supraglottic airway should be used. In addition, a supraglottic airway may be placed preferentially in lieu of face-mask ventilation after initial intubation attempts in any patient exhibiting predictors of difficult mask ventilation. Several supraglottic airways are commercially available. The decision to use one or the other of these devices depends on clinical experience

and availability of the devices, but at least one type of supraglottic airway should be included in difficult airway storage units. The devices have been described as either periglottic or retroglottic. Periglottic airways consist of a tube connected to an oval, somewhat pear-shaped mask with an inflatable or hydrocolloid gel-filled rim that fits into the periglottic area. The tip of the bowl of the mask, when correctly positioned, sits within the proximal end of the esophagus while the heel of the mask abuts the base of the tongue or within the vallecula (Fig. 38-14). Retroglottic airways have two inflatable cuffs: one distal designed to occlude the esophagus, and a more proximal cuff intended to seal the retropharynx. Between these two cuffs lies an opening through which air can flow (Fig. 38-15). Such devices have been used quite successfully to provide rescue ventilation in patients in whom face-mask ventilation has failed.⁵⁵ Although practice with supraglottic airways is recommended before use in an emergency, the insertion and ventilation success rate are high even in unskilled hands.^{56–58}

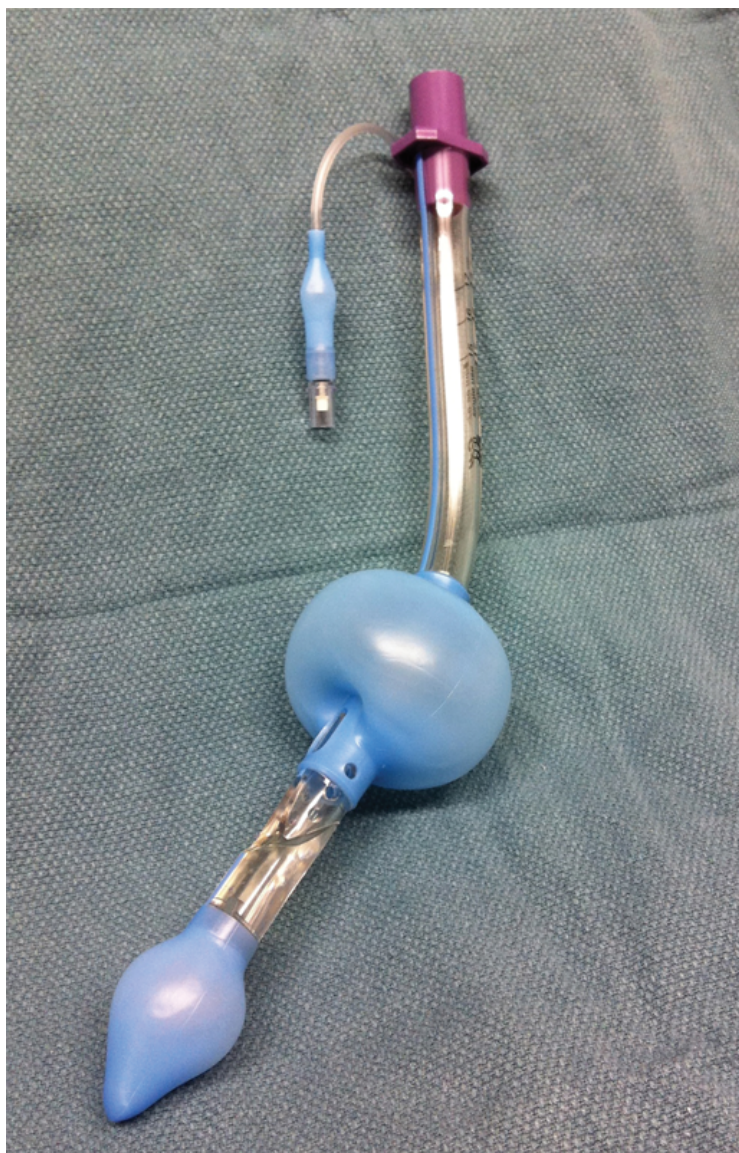
Figure 38-14



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Dorsal view of the laryngeal mask airway (LMA) showing position relative to pharyngeal anatomy. (Image courtesy LMA North America, Inc.)

Figure 38-15



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A retroglottic airway, the laryngeal tube. A single port inflates both the distal esophageal and more proximal pharyngeal balloons. The airway tube terminates in an opening just distal to the pharyngeal balloon. (Photograph courtesy A Joffe.)

The airway seal, that is to say, the airway pressure at which gas escapes around the cuff has a wide range depending on the device. The LMA-Classic and other similarly designed airways typically seal the airway up to pressures of 15 to 20 cm H₂O.⁵⁹ Devices such as the air-Q (Mercury Medical, Clearwater, FL) and the i-Gel (Intersurgical, Ltd., Berkshire, UK) supply a seal of approximately 25 cm H₂O and have the advantage of shorter, larger diameter airway tubes that allow direct passage of usual adult size tracheal tubes through them (7 to 8 mm inner diameter).^{60,61} The LMA-Proseal is the most extensively studied device with the most reliable airway seal, up to pressures of 40 cm H₂O, although the reported average is approximately 30 cm H₂O.⁶² It should be noted that the LMA-Classic and similar devices will not necessarily prevent aspiration of gastric contents, although this has not been a commonly reported problem during either elective use,⁶³ or during emergency ventilation. A number of supraglottic airways (LMA-Proseal, LMA-Supreme, i-Gel) have incorporated an esophageal vent to minimize the aspiration risk, although the benefit in this regard is contentious and there is a dearth of data supporting this claim.

Besides their simplicity, ease of insertion, and effectiveness, a major advantage of supraglottic airways is that they can provide a direct route to the larynx and trachea. When correctly seated in the supraglottis, the termination of the airway tube of the device sits directly opposite the glottic aperture. Intubation of the trachea through these devices has been described using several techniques. The most successful, however, involve the blind passage of an ETT through the device or fiber-optic bronchoscope aided techniques.^{64,65} The clear advantage of the latter technique is that it allows direct visual guidance and confirmation of intratracheal placement, and reported success rates are higher for fiber-optic versus blind intubation.^{45,66} The lumen of the originally designed, or “classic” size 3 LMA is only large enough to accept a 6-mm ETT, whereas the size 5 LMA will accept a 7-mm ETT. For this reason, difficult airways supplies for use in the intensive care unit may include newer supraglottic airways as detailed above so as to avoid this limitation.

An LMA specifically designed for intubation, the LMA-Fastrack, is modified to include a lumen large enough to accept an 8-mm ETT, a rigid structure to allow easier tube placement, and a bar to lift the epiglottis away from the laryngeal aperture. Like the classic LMA, placement and ventilation success rates are high with the intubating LMA, even for novice users.^{67,68} Intubation can be done blindly or using fiber-optic guidance; although blind intubation is usually successful, multiple attempts may be necessary.⁶⁸ Fiber-optic guidance results in a higher success rate on the first attempt.

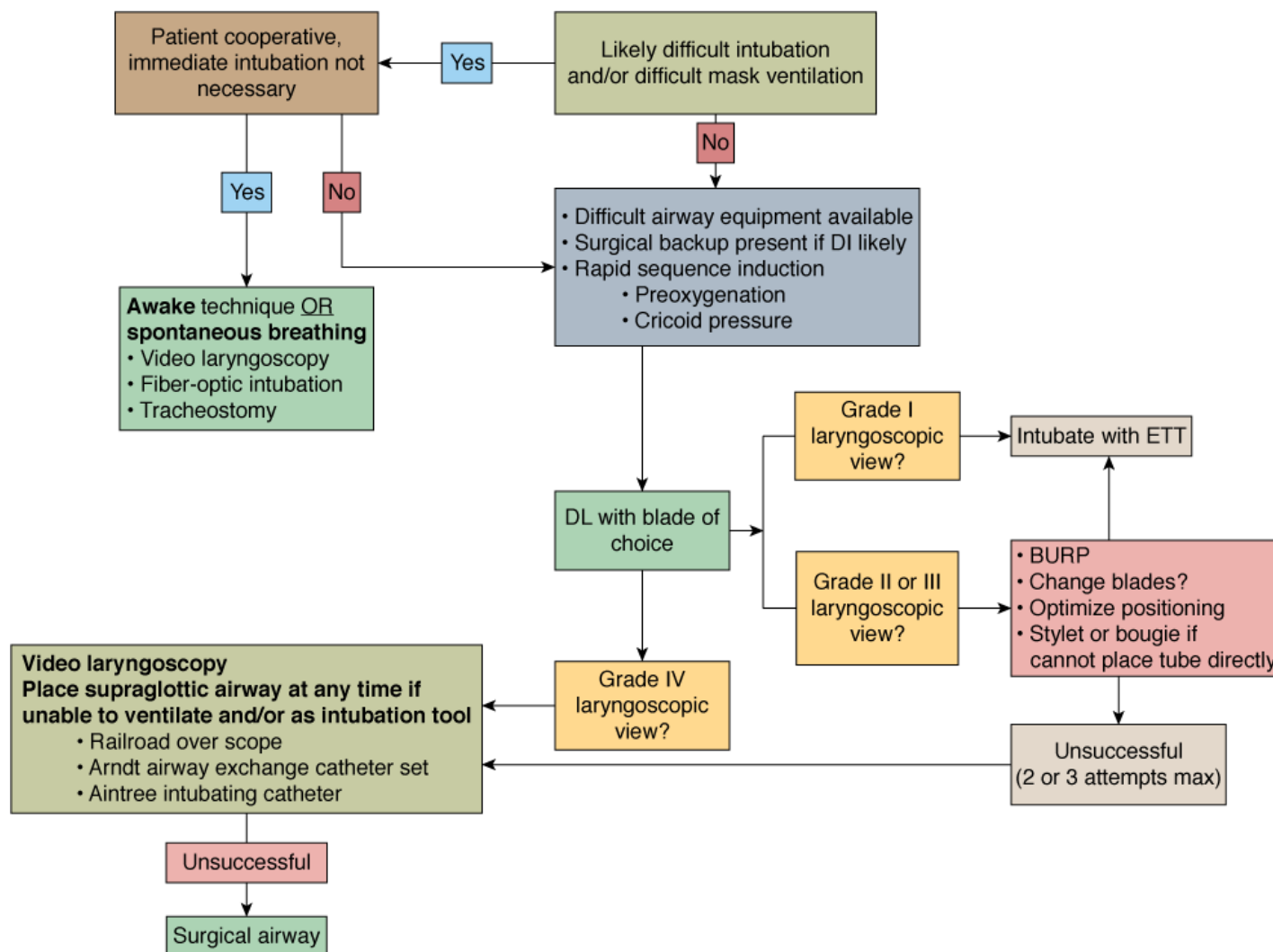
Failed Intubation and Ventilation: Cricothyrotomy

If intubation has been unsuccessful and ventilation via face mask and supraglottic airway is compromised, particularly if hypoxemia is present, more invasive airway management techniques must be employed. An important skill in airway management is the ability to recognize the need for a surgical airway *before* the development of injury-inducing, severe hypoxemia. The trachea can be accessed either percutaneously or surgically via the cricothyroid membrane; the approach used depends on the skill of the airway practitioner, and surgeon and equipment availability. Percutaneous placement of a large bore needle or catheter (13 to 16 gauge) permits transtracheal jet ventilation, which generally will provide adequate oxygenation and ventilation before a definitive surgical tracheotomy.⁶⁹ In order to carry out transtracheal jet ventilation, however, a high-pressure oxygen source and an adjustable, high-pressure device (jet injector) must be available; attempted manual ventilation through a catheter is largely ineffective. In addition, the potential complications associated with transtracheal jet ventilation are severe, particularly if the catheter is inadvertently positioned outside the trachea, and include barotrauma, arterial perforation, and gas trapping because of airway obstruction or insufficient expiratory time.⁷⁰ The correct position of the cannula should be verified by syringe aspiration of air before initiating ventilation, and care must be taken to avoid kinking of the catheter at the insertion site. For these reasons, transtracheal jet ventilation is becoming an increasingly uncommon recommendation in these situations and the necessary equipment has largely been relegated to the operating room.

Surgical cricothyrotomy is an effective and relatively safe and rapid way to secure the airway in experienced hands.^{71,72} The standard surgical cricothyrotomy technique utilizes a #11 scalpel to penetrate the skin, subcutaneous tissue, and cricothyroid membrane, followed by hemostat-aided dilation of the cricothyroid space before tube insertion. The “rapid four-step technique” uses a #20 scalpel to penetrate the cricothyroid membrane in one move, followed by caudal retraction of the cricoid cartilage and introduction of the tracheal tube.⁷³ The latter technique appears to be faster with a comparable complication rate. Ideally, these techniques should be reviewed and practiced on a model before use in the emergency setting.

Figure 38-16 is a difficult airway algorithm for emergency intubation situations.

Figure 38-16



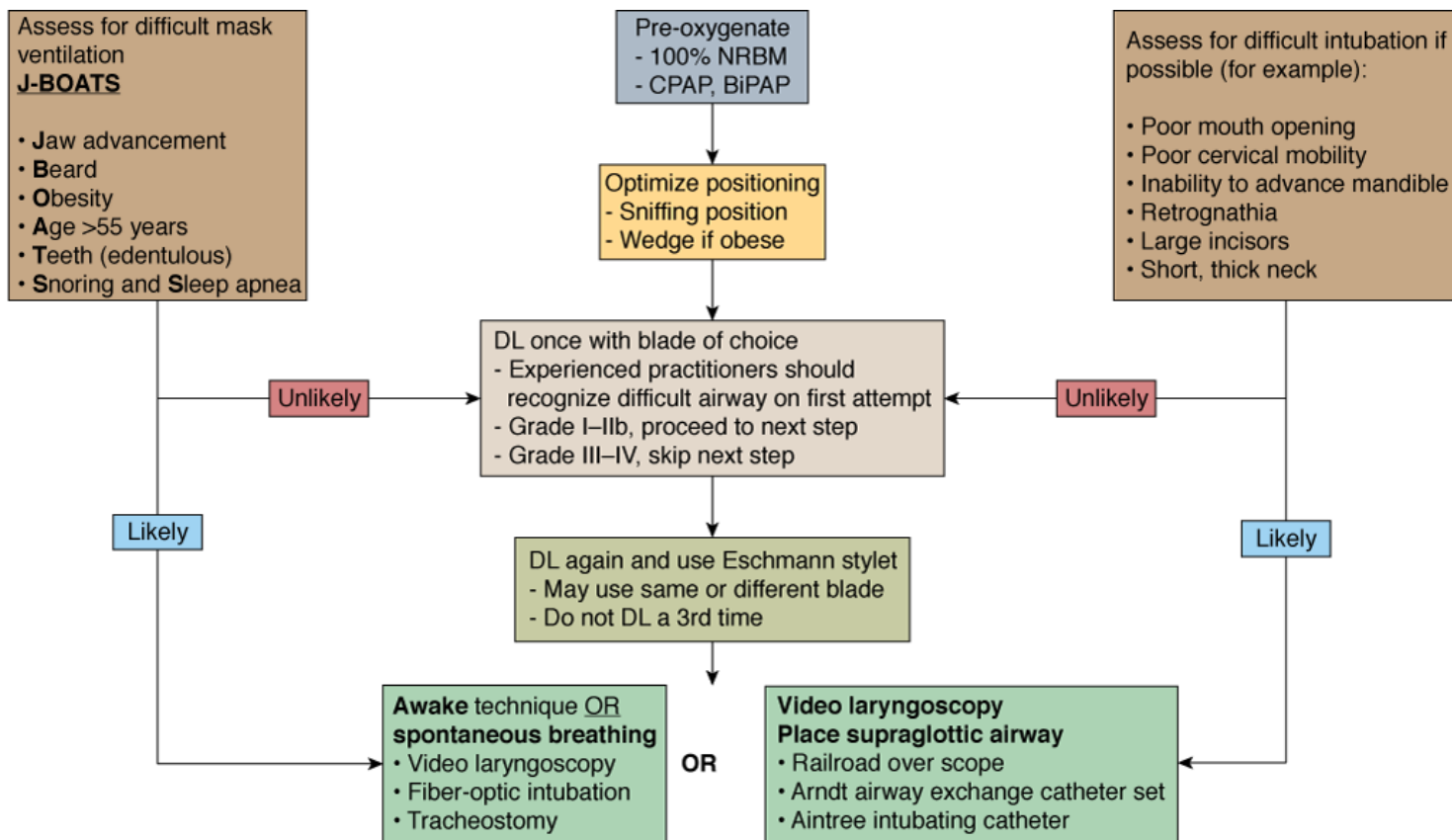
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Airway management algorithm for emergent tracheal intubations. Additional help should be sought at the first indication of a difficult airway situation. It is critical that movement to the next step in the algorithm not be prolonged by repeated attempts at direct laryngoscopy that may traumatize the airway and lead to failed ventilation and oxygenation. Confirmation of intratracheal placement should always use CO₂ detection combined with clinical findings. *DI*, difficult intubation; *ETT*, endotracheal tube.

Verification of Intratracheal Tube Placement and Position

A disastrous complication of endotracheal intubation is the unrecognized placement of the ETT in the esophagus.⁷⁴ Placement of the tube into a main stem bronchus is less likely to result in death or severe neurologic injury, but is still a risk for significant morbidity. Although misplacement might seem simple to avoid or to detect, even experienced clinicians are sometime fooled, especially following a difficult intubation (Fig. 38-17). At one time, airway management failures including unrecognized esophageal intubation were a small but significant cause of intraoperative mortality, morbidity, and liability claims. Minimum standards for monitoring established by the American Society of Anesthesiologists in 1986 included a directive that all tracheal intubations be confirmed by the measurement of carbon dioxide in the exhaled gas (capnometry). This standard, likely complemented by the additional standard that ventilation be continuously monitored by capnometry during general anesthesia, has been associated with a reduction in the incidence of respiratory-related anesthesiology legal claims, and a greater than 50% relative reduction in claims related to esophageal intubation.⁷⁵ It is assumed that the incidence of esophageal intubation has also decreased, although this is not directly proven.

Figure 38-17



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Incorporation of considerations for difficult ventilation and intubation. Help and airway equipment should be summoned early. When a “cannot intubate, cannot ventilate” situation occurs, or there is an inability to intubate by the means listed, the algorithm culminates in a surgical airway.

Confirmation of Intratracheal Placement

Clinical signs of intratracheal placement include direct observation of the tube entering the glottis, condensation in the ETT with exhalation, and observation and auscultation of the chest. Unfortunately, all clinical methods can and have failed. Inflation of the stomach results in some movement of the chest wall, creating sounds of air movement and some return of gas during the expiratory phase. Experimentally, placing a tube in the esophagus and attempting positive pressure ventilation actually results in some gas exchange because of diaphragmatic movement with consequent lung ventilation. This mechanism is thought to account for cases in which a patient survives for many minutes following an intubation that eventually proves to be esophageal. This is clearly the exception, however, and esophageal intubation that is unrecognized for more than a few minutes generally leads to adverse consequences.

Although observation of the tube entering the glottis would seem a fail-safe test, there are numerous reported cases of esophageal intubation in which the clinician stated that the cords were seen. This may occur because the tube itself obscures the view during placement and the tube may then enter the esophagus.

Multiple methods should be used to confirm intratracheal tube placement for each intubation attempt. These methods should include capnometry, which is now considered the “gold standard” for confirmation of intratracheal placement. Capnometry may fail to detect esophageal intubation if intubation follows a period of mask ventilation because of exhaled gas that has entered the stomach,^{76–78} or if carbonated beverages are ingested shortly before intubation.⁷⁹ In these instances, however, the “exhaled” CO₂ levels will decrease to near zero as the gastric gas becomes diluted, whereas CO₂ will persist if the ETT is in the trachea. False negatives can occur if there is no CO₂ production or pulmonary blood flow, for example during cardiac arrest or after massive pulmonary embolism,⁸⁰ or after marked hyperventilation.⁸¹

Capnography, which refers to the graphical description of the exhaled CO₂ waveform, has become standard in operating rooms in the United States and much of the rest of the world. Portability and cost has limited the use of capnography in areas where tracheal intubation is performed less frequently. Alternatives to capnography include portable capnometry using hand-held devices, or colorimetric CO₂ detectors. The latter contain an indicator (metacresol purple) that changes color from purple to yellow when it is exposed to CO₂. Colorimetric capnometry is comparable to capnography in differentiating tracheal from esophageal intubation.^{82,83}

An alternative to exhaled CO₂ detection is to apply gentle suction to the in situ tube; if the tube is in the esophagus, the suction causes the pliable esophageal tissue to occlude the tube and no air can be aspirated. If the tube is in trachea, however, the semirigid walls prevent tube occlusion when suction is applied, and air can be obtained. Suction can be applied manually with a syringe, or more commonly with a self-inflating bulb (“bulb syringe”). This technique has good sensitivity for detection of esophageal intubation in elective and emergency settings,^{84–88} but failures have been reported, particularly in pregnant subjects.⁸⁹ In addition, this technique has a relatively low specificity in detecting esophageal intubation compared to CO₂ detection, and incorrect identification of correct tube placement (bulb does not reinflate or air cannot be aspirated) has been reported in patients with morbid obesity, obstructive lung disease, copious secretions, and endobronchial intubation.^{86,90,91} Thus, this technique cannot be advocated as a replacement for CO₂ detection, although it appears to have particular value in patients where CO₂ detection may fail, particularly during cardiac arrest.⁸⁶

Assessing Depth of Intratracheal Placement

The ETT tube can move as much as 5 cm during maximal cervical range of motion from flexion through extension; the tube moves cephalad as the neck extends.⁹² The average distance from larynx to tracheal carina is 12 to 14 cm, and the top of the cuff of tracheal tubes in common use begins approximately 6 cm above the tip of the tube. Thus, the ETT should ideally be placed 5 ± 2 cm above the carina to ensure that the tip does not migrate endobronchially during neck flexion, or that the cuff does not herniate through the glottis with cervical extension.⁹²

Endobronchial placement can result in atelectasis and hypoxemia,⁹³ and inadequate depth of placement has several possible adverse consequences. If the cuff is between the cords the seal will be inadequate risking both aspiration and inadequate ventilation. Adding more air to the cuff usually will not improve the seal as the cuff has a circular profile and the pentagonal shape of the glottic opening leaves the corners open. Inadequate depth of placement also may lead to serious long-term problems with subglottic stenosis. If the cuff is overinflated in the immediate subglottic region, mucosal ischemia may result. As this heals, scarring will lead to narrowing in the region of the cricoid. This problem is extremely difficult and is not readily surgically correctible.

Although auscultation of breath sounds is a reasonable adjunctive test for confirmation of ETT depth, the incidence of missed endobronchial intubation using auscultation alone may approach 10%.⁹³ In addition, auscultation does not detect ETTs that are either dangerously close to the carina, or positioned too high in the trachea.

The risk of endobronchial intubation can be minimized by: (a) observation of the ETT as it passes through the glottis; insertion of the tube no more than 3 to 4 cm after the upper end of the ETT cuff passes the glottis; (b) insertion of the tube no deeper than 20 cm at the teeth in adult females and 22 cm in adult males;⁹⁴ (c) palpation of the ETT cuff in the sternal notch.⁹⁵ These benchmarks will result in good placement in a high proportion of adults, but may fail at the extremes of stature, and are more likely to fail in women.⁹⁶

Other methods for detecting endobronchial placement include chest radiography and fiber-optic bronchoscopy. Chest radiography confirms the position of the tube relative to the carina. Even if the carina is not visible, a tube tip overlying the third or fourth thoracic vertebral body with the head in the neutral position means the tube is in good position.⁹² Fiber-optic bronchoscopy is as reliable as chest radiography and may be less expensive, but is not readily available at all intubation sites. Chest radiography has the added advantage of providing information about the lung fields as well.

Pharmacologic Adjuncts to Tracheal Intubation

Except in cases of cardiac arrest, coma, or extreme neuromuscular weakness, tracheal intubation is facilitated by administration of pharmacologic agents, particularly if direct laryngoscopy is used. The goals of the administered drugs are threefold: (a) to provide optimal intubating conditions by minimizing muscular tone; (b) to ensure patient comfort during the procedure; and (c) to minimize physiologic responses to intubation, as follows.

The upper airway is richly innervated, resulting in a series of physiologic responses during intubation, including activation of both the sympathetic and parasympathetic nervous systems.⁹⁷ Bradycardia occasionally results, most commonly in children. In unanesthetized adults, blood pressure following intubation may reach levels as high as 250/150 mm Hg and the pulse will frequently rise to near maximal heart rates. Appropriate anesthesia will generally prevent these responses.

Direct laryngoscopy in the unanesthetized patient stimulates a strong gag reflex, and likewise insertion of a tube in the trachea will stimulate coughing. These reflexes are not only noxious for the patient, but can make tracheal intubation more challenging. Normal subjects develop a mild, clinically insignificant increase in lower-airway resistance following tracheal intubation,⁹⁸ and patients with reactive airways may develop marked bronchoconstriction. Bronchospasm following induction of general anesthesia accounts for a small but significant proportion of perioperative morbidity; notably, many cases of severe intraoperative bronchospasm occur in patients without previously diagnosed reactive airways disease.^{74,99–101} Ideally,

tracheal intubation of patients with known asthma or other reactive airways disease should be preceded by administration of inhaled bronchodilators, in addition to anesthetic drugs with bronchodilating properties (e.g., propofol or ketamine, as discussed below).

Direct laryngoscopy causes mechanical obstruction of cerebral venous flow, and increases cerebral blood flow related to increased cerebral metabolic rate and transient increases in P_{CO_2} .^{102,103} In the presence of preexisting intracranial pathology, these factors can lead to a rapid rise in intracranial pressure and carry a risk of herniation of brain contents. The increase in intracranial pressure can be blunted by the prior administration of anesthetic agents.

In addition to topical anesthetics (discussed earlier), the typical drugs administered to facilitate tracheal intubation are sedative-hypnotics, intravenous [lidocaine](#), and neuromuscular blocking drugs (“muscle relaxants”).

Sedative-Hypnotics

The most common sedative/hypnotics used as adjuncts to tracheal intubation are the short-acting barbiturates, propofol, etomidate, ketamine, and the benzodiazepines. With the exception of ketamine, these drugs are all γ -amino butyrate receptor agonists. They all have dose-dependent cardiovascular and/or respiratory depressant effects of varying degrees, and will result in loss of airway control at higher doses. Barbiturates, propofol, and etomidate are all very effective at reducing intracranial pressure; benzodiazepines are less potent in this regard, and ketamine can increase intracranial pressure.

Short-Acting Barbiturates

Sodium thiopental, the prototypical intravenous induction agent, is highly lipid soluble and thus rapidly enters the brain to produce unconsciousness. The return of consciousness is also rapid because of rapid redistribution of the drug. There is marked respiratory depression following injection with transient apnea the rule. Barbiturates cause vasodilation and myocardial depression, making them a poor choice for the hypovolemic patient. The typical induction dose of sodium thiopental is 3 to 5 mg/kg intravenously, with reduced doses given in the presence of cardiovascular instability, hypovolemia, and advanced age. Higher doses may be necessary if there is a history of heavy [alcohol](#) or tranquilizer use.

Propofol

Propofol produces rapid loss of consciousness after bolus intravenous administration, with recovery beginning within 1 to 2 minutes. Propofol has cardiovascular depressant effects that are similar in magnitude to those seen with barbiturates, and its potency appears to be increased during hemorrhagic shock.¹⁰⁴ Thus, large doses of propofol should be avoided in patients with hypovolemia or significant cardiovascular depression for any reason. Propofol profoundly suppresses respiratory drive and airway reflexes, and has bronchodilating properties; thus, it is a useful agent for intubation of patients without concomitant administration of a neuromuscular blocking drug, or for intubation of patients with bronchospasm or reactive airways.¹⁰⁵ Propofol rarely causes myoclonus after bolus administration. Propofol is mixed in a lipid-based solution that supports bacterial growth; thus, propofol-containing syringes must be handled carefully and administered within a few hours of opening. Typical induction doses of propofol are 1 to 2.5 mg/kg intravenously, depending on age and cardiovascular stability.

Etomidate

Etomidate has a rapid onset of action and a rapid recovery, has little effect on myocardial function, and has mild respiratory depressant effects. In addition, it has mild $\alpha_2\text{-}\beta$ -adrenergic receptor agonist activity, which contributes to its lack of hypotensive effects compared to other induction agents.¹⁰⁶ Because of these favorable properties, etomidate has long been favored as an adjunct to tracheal intubation. Etomidate, however, can transiently suppress adrenal function by reversibly inhibiting 11 β -hydroxylase, and prolonged infusions of etomidate in critically ill patients have been associated with severe suppression of adrenal cortical function and increased mortality.¹⁰⁷ Moreover, recent data suggest that even single doses of etomidate can result in prolonged adrenal suppression in critically ill patients, particularly those with sepsis.^{108–110} This observation has generated considerable concern and debate about the advisability of using etomidate as an adjunct to tracheal intubation in critically ill patients, with some experts calling for caution if not complete abandonment of the use of etomidate in this patient population.^{111,112} It remains unclear, however, if etomidate exerts an independent effect on mortality,¹¹³ and a randomized, controlled trial comparing etomidate to ketamine for rapid sequence induction in acutely ill patients revealed no mortality difference between the two groups.¹¹⁴ Further study of this issue will be necessary to definitively determine if etomidate should continue to be used as an intubation adjunct in critically ill patients; in the interim, it is reasonable to consider the use of alternative agents where feasible.

Etomidate also causes self-limited myoclonus relatively frequently. The latter may not be seen if etomidate is given in conjunction with a neuromuscular blocking drug. In addition, etomidate has proemetic effects and can cause a burning sensation when injected through a peripheral venous catheter, and these latter effects limit its use as an elective induction agent. The typical induction dose is 0.1 to 0.3 mg/kg intravenously; doses of 4 to 8 mg in an adult patient will typically produce deep sedation with maintenance of spontaneous ventilation.

Ketamine

Ketamine is an *N*-methyl-d-aspartate-receptor antagonist, which gives it a different sedation and side effect profile in comparison with the other agents discussed here. Ketamine produces rapid loss of consciousness, although patients often appear awake because their eyes often remain open. Recovery from an intravenous dose of 2 mg/kg may take 10 to 15 minutes. Ketamine is a direct myocardial depressant^{115,116} but clinically produces cardiovascular stimulation via release of catecholamines.¹¹⁷ This sympathomimetic effect also gives ketamine bronchodilating properties.^{118,119} Ketamine's major adverse effects include the potential for myocardial ischemia resulting from cardiovascular stimulation, increased intracranial pressure, and unpleasant dreams or hallucinations during the recovery phase. As noted above, ketamine appears to be equivalent if not superior to etomidate in safety when used for rapid sequence induction in acutely ill patients.¹¹⁴ The typical induction dose of ketamine is 0.5 to 1 mg/kg, intravenously.

Benzodiazepines

Midazolam is the benzodiazepine most widely used for hypnosis during intubation because of its rapid onset and short duration of action. It and the other benzodiazepines are more suited as sedative agents rather than for full induction of anesthesia, as the large doses of these drugs necessary to produce loss of consciousness result in prolonged sedation. In critically ill patients, however, smaller doses can result in loss of the airway, respiratory depression, and apnea; thus, titrated and closely monitored administration is recommended. Typical doses of midazolam for sedation during awake intubation are 0.5 to 2 mg intravenously.

Intravenous Lidocaine

High-dose intravenous lidocaine has general anesthetic effects, including a reduction of cerebral metabolic rate and a blunting of the hemodynamic response to intubation. Lidocaine 1.5 mg/kg given intravenously (before laryngoscopy) blunts the increase in intracranial pressure typically seen during intubation of patients with intracranial pathology.¹²⁰ Intravenous lidocaine also suppresses cough and bronchoconstriction during laryngoscopy, but only when given at doses of 1.5 mg/kg or greater and approximately 3 minutes before intubation.¹²¹

Neuromuscular Blocking Drugs

Although there are a wide variety of neuromuscular blocking drugs available, only two, succinylcholine and rocuronium, are sufficiently rapid in onset to be of value for emergency tracheal intubation. Succinylcholine's pharmacokinetic profile makes it the ideal drug for emergency tracheal intubation. Succinylcholine binds to the nicotinic receptor at the neuromuscular junction to cause depolarization of the end plate and, eventually, of surrounding fibers. This effect persists for several minutes, thus causing the muscle membrane to remain depolarized and refractory to further impulses. The diffuse depolarization of the muscles can be seen as generalized fasciculations. The usual intubating dose of 1 mg/kg of succinylcholine results in adequate intubating conditions in 90% of patients at 1 minute. The termination of drug action results from the metabolism of circulating succinylcholine by pseudocholinesterase with return of muscle strength in 3 to 5 minutes in most patients. Pseudocholinesterase deficiency occurs rarely, and results in prolonged action of succinylcholine.

Succinylcholine has a number of side effects, some of which are potentially lethal. Succinylcholine is a trigger agent for malignant hyperthermia, and should not be administered to malignant hyperthermia-susceptible patients. Administration of succinylcholine results in an efflux of intracellular potassium resulting in a clinically insignificant transient rise in serum potassium in most patients. In patients with central neurologic deficits, however, such as stroke, traumatic brain injury, encephalopathy, or spinal cord injury, or after peripheral denervation, succinylcholine can lead to massive potassium release and potassium levels over 10 mEq/L.^{122,123} Similar responses are seen in burn patients beginning several days after the burn, in patients with myopathies, rhabdomyolysis, and after crush injuries.¹²² The mechanism of hyperkalemia is in part related to extrajunctional proliferation of nicotinic receptors, but may also be related to direct muscle membrane damage during critical illness. The reported mortality with succinylcholine-induced hyperkalemia is more than 10%.¹²² Succinylcholine is thus absolutely contraindicated more than several days to a week following a burn or new neurologic deficit. In addition, given the high incidence of critical illness-induced myopathy and neuropathy, succinylcholine should be avoided in patients who have been critically ill for a week or more.¹²⁴ Given the potential lethality of the hyperkalemic response after succinylcholine, an alternate agent should be used if there is any potential contraindication to succinylcholine.

Succinylcholine may also result in a transient increase in intracranial pressure, although this is probably not clinically significant and should not countermand the use of succinylcholine if rapid control of the airway is necessary.¹²⁵

Rocuronium is a competitive antagonist of acetylcholine at the neuromuscular junction. Rocuronium is "nondepolarizing," does not result in muscle fasciculation or potassium release, and does not act as a trigger for malignant hyperthermia. Rocuronium at an intubating dose of approximately 1 mg/kg results in acceptable intubating conditions at 1 minute after administration, although intubation conditions are generally inferior to those achieved after succinylcholine.¹²⁶ Because rocuronium does not cause muscle fasciculation, which, in turn, can increase oxygen consumption, it may

allow a longer apneic period before oxyhemoglobin desaturation compared to succinylcholine.¹²⁷ Rocuronium's major drawback is that its duration of action is determined by hepatic and renal clearance, and has a variable range of 30 to 90 minutes. This makes rocuronium less than ideal for patients with difficult airways, as the option of allowing return of spontaneous ventilation should intubation and face-mask ventilation fail is not available, as it is for succinylcholine. Nonetheless, rocuronium is a reasonable alternative to succinylcholine for rapid sequence induction, particularly in patients who are at risk for complications from the latter agent.

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